

T.POOJITHA (192412318)

ITA0609

Exp-8 : Write a program to implement Linear Regression (LR) algorithm in python

Aim

To implement the Linear Regression (LR) algorithm using Python and predict the output based on the given input data.

Program Logic

1. Import the LinearRegression class.
2. Create input (X) and output (Y) datasets.
3. Create the Linear Regression model.
4. Train the model using fit().
5. Predict the output for a new input value (6).
6. Display the predicted value, slope, and intercept.

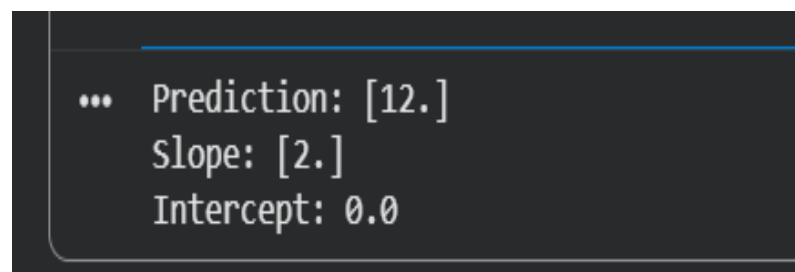
program

```
from sklearn.linear_model import LinearRegression
X=[[1],[2],[3],[4],[5]]
Y=[2,4,6,8,10]
model=LinearRegression()
model.fit(X,Y)
pred=model.predict([[6]])
print("Prediction:",pred)
print("Slope:",model.coef_)
print("Intercept:",model.intercept_)
```

sample input

```
X = [[1],[2],[3],[4],[5]]
Y = [2,4,6,8,10]
```

output



```
... Prediction: [12.]
    Slope: [2.]
    Intercept: 0.0
```

Result:

The Linear Regression model was successfully implemented, and it correctly predicted the output value **12** for the input **6**.

